

Business Problem Statement

(Student Performance Analytics Dashboard)

An engineering institution aims to better understand students' academic performance, skill development, internship participation, and placement outcomes in order to improve overall student success and employability. The management team has observed variations in academic achievement, attendance patterns, certification completion, internship experiences, and placement performance across departments, admission categories, and student demographics.

They are particularly interested in identifying which factors, such as admission type, attendance, previous academic records, certifications, internships, projects, and aptitude performance, contribute most significantly to student success and placement readiness.

You are tasked with analyzing the Student Performance Analytics dataset to answer the following overarching business question:

"How can the institution leverage student data to identify academic trends, enhance student engagement, improve placement readiness, and support data-driven decision-making for academic and career development?"

Deliverables

1. Data Preparation & Modeling (Python)

Clean, preprocess, and transform the raw student dataset for analysis by handling missing values, duplicate records, inconsistent categories, and outliers.

2. Exploratory Data Analysis (Python)

Perform exploratory data analysis to uncover relationships among attendance, academic performance, certifications, internships, project domains, and placement outcomes.

3. Data Analysis (SQL)

Load the cleaned dataset into a relational database, structure the data efficiently, and execute analytical queries to derive insights related to student performance, placement trends, academic risk, and employability factors.

4. Visualization & Insights (Power BI)

Develop interactive dashboards that provide insights into student demographics, academic performance, skills development, internship participation, placement analytics, and career readiness.

5. Report and Presentation

Prepare a comprehensive report summarizing key findings, trends, and recommendations. Create a presentation to communicate insights effectively to academic administrators and stakeholders.

6. GitHub Repository

Maintain a well-structured GitHub repository containing Python notebooks, SQL scripts, Power BI dashboard files, datasets, documentation, and project deliverables.

Expected Outcomes

- Identify factors influencing student academic performance.
- Analyze the relationship between attendance and CGPA/SGPA.
- Evaluate the impact of certifications and internships on placement opportunities.
- Study admission trends and their effect on student outcomes.
- Detect academically at-risk students for early intervention.
- Understand placement patterns across departments and domains.
- Support institutional decision-making through actionable insights and KPI monitoring.